

SQL Views

- A SQL view is a SELECT statement that is stored as a database object.
- A SQL view acts as a virtual table, but contains no data.
- You can use a view anywhere you would use a table including in a SELECT, INSERT, UPDATE, or DELETE statement.

Benefits of Using Views

- **Design Flexibility:** By using a view instead of a query in an application, it is easier to make changes to the underlying table structure.
- **Improved Security:** By using a view to return data from tables instead of a SELECT, you can hide the WHERE clause or other columns to which you do not want the user to have access.
- **Query Simplification:** You can write simple select statements against views, which handle complex queries and joins.

View Example

```
USE WORLD;  
  
CREATE VIEW city_country AS  
SELECT ci.name AS city_name, co.name AS country_name  
FROM city ci  
      JOIN country co  
      ON ci.CountryCode = co.Code;
```

CREATE VIEW city_country AS

- Create a new VIEW object and give it the name city_country
- The AS statement precedes the query that will be assigned to the VIEW

SELECT ci.name AS city_name, co.name AS country_name

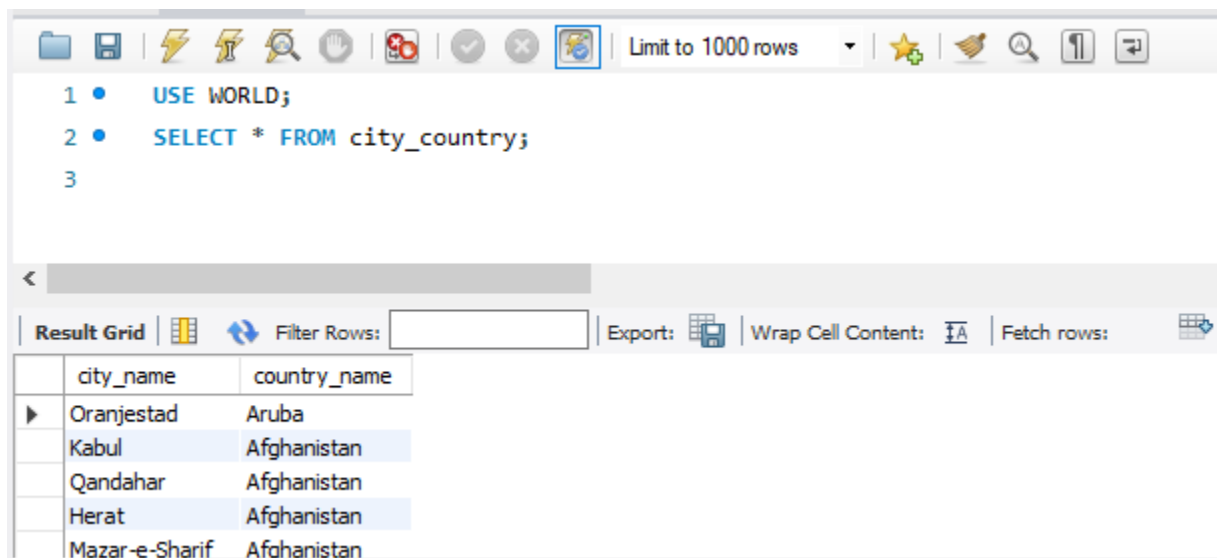
- Only the columns defined in the SELECT statement will be available to the VIEW
- It is a good idea to provide column alias in the select because the VIEW will not have access to the underlying table structure.

FROM city ci

JOIN country co

ON ci.CountryCode = co.Code;

- The JOIN statement of the SELECT.
- Once you have created a VIEW, you can run SQL statements using the VIEW as if it were a table.



- By creating a VIEW, we can run selects that retrieve data from multiple tables without having to re-code a join.
- Notice how the SELECT * retrieves only the rows defined in the SELECT statement used in the VIEW creation.
- If you want to drop a VIEW, we can run the DROP VIEW statement
- If you want to modify an existing view you can use the statement CREATE OR REPLACE VIEW. That way you do not have to run a DROP VIEW statement and then a CREATE VIEW statement.

Creating Views that Can Be Used with an UPDATE statement

- There are some restrictions to creating a VIEW if you want to be able to run an UPDATE statement against it.
 - SELECT list cannot include a DISTINCT clause.
 - SELECT list cannot contain aggregate functions (SUM, COUNT, MIN, MAX, AVG, COUNT(*))
 - SELECT statement cannot use GROUP BY or HAVING.
 - The VIEW cannot include a UNION operator.
- If you use any of the restricted statements, your view will be read only.